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SEMICONDUCTOR MEMORY DEVICE AND METHOD OF FABRICATING THE SAME

Abstract

Disclosed is a semiconductor memory device including a lower structure and an upper structure on the lower structure. The upper structure includes a first substrate, an upper wiring line on the first substrate, a lower power line in a lower portion of the first substrate, and a first bonding pad between the lower power line and the lower structure. The lower power line and the first bonding pad are electrically connected to each other. The upper structure is electrically connected through the first bonding pad to the lower structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This U.S. nonprovisional application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032132 filed on Mar. 6, 2024, in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The present inventive concepts relate to a semiconductor memory device, and more particularly, to a semiconductor memory device including vertical channel transistors and a method of fabricating the same.

[0003] With the trend of reduction in design rule of semiconductor memory devices, the fabrication technology is being improved to increase integration, operating speed, and yield of the semiconductor memory device. Accordingly, transistors with vertical channels have been suggested to increase their integration, resistance, current driving capability, and other characteristics.

SUMMARY

[0004] Some embodiments of the present inventive concepts provide a semiconductor memory device having improved electrical properties and increased integration.

[0005] Embodiments of the present inventive concepts are not limited to those mentioned above, and other embodiments, which have not been mentioned above, will be clearly understood to those skilled in the art from the following description.

[0006] According to some embodiments of the present inventive concepts, a semiconductor memory device may comprise: a lower structure; and an upper structure on the lower structure. The upper structure may include: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a first bonding pad between the lower power line and the lower structure. The lower power line and the first bonding pad may be electrically connected to each other. The upper structure may be electrically connected through the first bonding pad to the lower structure.

[0007] According to some embodiments of the present inventive concepts, a semiconductor memory device may comprise: a lower structure; and an upper structure on the lower structure. The upper structure may include: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a first bonding pad between the lower power line and the lower structure. The lower structure may include: a second substrate that includes a cell region; and a second bonding pad on the cell region. The first bonding pad may be exposed through a rear surface of the upper structure. The second bonding pad may be exposed through a front surface of the lower structure. The front surface of the lower structure may be in contact with the rear surface of the upper structure. The first bonding pad and the second bonding pad may directly contact each other.

[0008] According to some embodiments of the present inventive concepts, a semiconductor memory device may comprise: a lower structure; an upper structure on the lower structure; and a bonding structure between the lower structure and the upper structure. The upper structure may include: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a backside through via that extends through the first substrate and connects the upper wiring line to the lower power line. The lower structure may include: a second substrate that includes a cell region; and a lower wiring layer on the cell region. The bonding structure may include: a first part in a lower portion of the upper structure; and a second part in an upper portion of the lower structure. The bonding structure may electrically connect the lower power line and the lower wiring layer to each other.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a block diagram illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0010] FIG. **2** is a simplified perspective view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0011] FIG. **3**A is a plan view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0012] FIG. **3**B is a plan view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0013] FIG. **4** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0014] FIGS. **5**A to **5**E and **6**A to **6**D are cross-sectional diagrams illustrating a method of fabricating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0015] FIG. **7** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0016] FIGS. **8**A, **8**B, **9**A, **9**B, **10**, **11**, and **12** are cross sectional diagrams illustrating a method of fabricating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0017] FIG. **13** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

DETAILED DESCRIPTION OF EMBODIMENTS

[0018] The following will now describe some embodiments of the present inventive concepts with reference to the accompanying drawings. Like reference numerals may indicate like components throughout the description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be understood that, although the terms “first,” “second,” “upper portion,” “lower portion,” etc. may be used herein to describe various elements or components, these elements or components should not be limited by these terms. These terms are only used to distinguish one element or component from another element or component. Therefore, a first element or component discussed below could be termed a second element or component. It is noted that aspects described with respect to one embodiment may be incorporated in different embodiments although not specifically described relative thereto. That is, all embodiments and/or features of any embodiments can be combined in any way and/or combination.

[0019] FIG. **1** is a block diagram illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.

[0020] Referring to FIG. **1**, a semiconductor memory device may include a memory cell array **1**, a row decoder **2**, a sense amplifier **3**, a column decoder **4**, and a control logic **5**.

[0021] The memory cell array **1** may include a plurality of memory cells MC that are arranged two-dimensionally or three-dimensionally. Each of the memory cells MC may be connected between a word line WL and a bit line BL that intersect each other.

[0022] Each of the memory cells MC may include a selection element TR and a data storage element DS, and the selection element TR and the data storage element DS may be electrically connected in series to each other. The selection element TR may be connected between the data

storage element DS and the word line WL, and the data storage element DS may be connected through the selection element TR to the bit line BL. The selection element TR may be a field effect transistor (FET), and the data storage element DS may be a capacitor, a magnetic tunnel junction pattern, or a variable resistor. For example, the selection element TR may include a transistor, a gate electrode of the transistor may be connected to the word line WL, and source/drain terminals of the transistor may be connected to the bit line BL and the data storage element DS.
[0023] The row decoder **2** may decode an address that is externally input, and may select one of the word lines WL of the memory cell array **1**. The address that is decoded in the row decoder **2** may be provided to a row driver (not shown), and in response to a control operation of control circuits, the row driver may provide a certain voltage to a selected word line WL and each of non-selected word lines WL.
[0024] In response to an address that is decoded from the column decoder **4**, the sense amplifier **3** may detect and amplify a voltage difference between a selected bit line BL and a reference bit line, and may then output the amplified voltage difference.
[0025] The column decoder **4** may provide a data delivery pathway between the sense amplifier **3** and an external device (e.g., a memory controller). The column decoder **4** may decode an address that is externally input and may select one of the bit lines BL.
[0026] The control logic **5** may generate control signals that control operations to write data to the memory cell array **1** and/or to read data from the memory cell array **1**.
[0027] FIG. **2** is a simplified perspective view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.
[0028] Referring to FIG. **2**, a semiconductor memory device may include a peripheral circuit structure PS on a semiconductor substrate **100** and may also include a cell array structure CS on the peripheral circuit structure PS. For example, the peripheral circuit structure PS may be positioned between the semiconductor substrate **100** and the cell array structure CS in a third direction D**3** perpendicular to a top surface of the semiconductor substrate **100**. In this description, the third direction D**3** may be called a vertical direction.
[0029] The peripheral circuit structure PS may include core/peripheral circuits formed on the semiconductor substrate **100**. The core/peripheral circuits may include a row decoder, a column decoder, a sense amplifier, and a control logic that are discussed in FIG. **1**.
[0030] The cell array structure CS may include bit lines BL, word lines WL, and the memory cells MC of FIG. **1** between the bit lines BL and the word lines WL. The memory cells MC of FIG. **1** may be arranged two-dimensionally or three-dimensionally on a plane defined by a first direction D**1** and a second direction D**2** that intersect each other. Each of the memory cells MC of FIG. **1** may include a selection element TR and a data storage element DS.
[0031] For example, the selection element TR may include a vertical channel transistor (VCT). A channel of the vertical channel transistor may have a shape that extends in a direction (e.g., the third direction D**3**) perpendicular to the top surface of the semiconductor substrate **100**. The data storage element DS may be a capacitor.
[0032] The present inventive concepts, however, are not limited thereto. The semiconductor memory device may include the cell array structure CS on the semiconductor substrate **100**, and may also include the peripheral circuit structure PS on the cell array structure CS. For example, the cell array structure CS may be positioned between the semiconductor substrate **100** and the peripheral circuit structure PS.
[0033] FIGS. **3**A and **3**B are plan views illustrating a semiconductor memory device according to some embodiments of the present inventive concepts. FIG. **4** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts.
[0034] Referring to FIGS. **3**A, **3**B, and **4**, a semiconductor memory device **1000** may include a lower structure LS and an upper structure US. The upper structure US may be provided on the lower structure LS. A rear surface USa of the upper structure US may be in contact with a front

surface LSa of the lower structure LS.

[0035] The upper structure US will be further discussed in detail with reference to FIGS. **3**B and **4**.

[0036] The upper structure US may include a semiconductor substrate **100** (referred to hereinafter as a first substrate). The first substrate **100** may be a silicon substrate, but the present inventive concepts are not limited thereto. A cell region and a peripheral region may be provided on the first substrate **100**. For example, the first substrate **100** may be provided thereon with a peripheral region R including transistors TR. The peripheral region R may be an area from which signals are provided through the transistors to the cell region.

[0037] An active pattern AP may be positioned on the first substrate **100**. The active pattern AP may be defined by an isolation pattern STI. The isolation pattern STI may include a dielectric material. The transistor TR and a source/drain pattern SD may be provided on the active pattern AP. The source/drain patterns SD may be correspondingly provided on opposite sides of the transistor TR. The source/drain patterns SD may be provided in an upper portion of the active pattern AP.

[0038] The first substrate **100** may include a deep device isolation pattern DTI that vertically (i.e., D**3** direction) penetrates or extends through the first substrate **100**. The deep device isolation pattern DTI may define the cell region or the peripheral region R. A width in a first direction D**1** of the deep device isolation pattern DTI may decrease with increasing distance in a vertical direction D**3** from a top surface of the first substrate **100** as shown in FIG. **4**.

[0039] The deep device isolation pattern DTI may penetrate the first substrate **100**. A top surface of the deep device isolation pattern DTI may be coplanar with that of the first substrate **100**. A bottom surface of the deep device isolation pattern DTI may be coplanar with that of the first substrate **100**. The deep device isolation pattern DTI may include a dielectric material.

[0040] A first interlayer dielectric layer **110** may be provided on the first substrate **100**. An upper wiring layer **111** may be provided in the first interlayer dielectric layer **110**. The upper wiring layer **111** may include active contacts AC, an upper wiring line W**1**, and an upper via VI. The active contact AC may be electrically connected to the source/drain pattern SD. The active contact AC may electrically connect the source/drain pattern SD and the upper wiring line W**1** to each other. The active contacts AC, the upper wiring line W**1**, and the upper via VI may each include a metallic material.

[0041] A second interlayer dielectric layer **120**, a third interlayer dielectric layer **130**, and a fourth interlayer dielectric layer **140** may be sequentially provided on the upper wiring layer **111**. For example, the first, second, third, and fourth interlayer dielectric layers **110**, **120**, **130**, and **140** may include a silicon oxide layer. For another example, one or more of the second, third, and fourth interlayer dielectric layers **120**, **130**, and **140** may be omitted. For another example, at least one interlayer dielectric layer may be formed on the fourth interlayer dielectric layer **140**.

[0042] A first metal layer M**1** may be provided in the second interlayer dielectric layer **120**. The first metal layer M**1** may include first wiring lines. According to some embodiments of the present inventive concepts, a lower substrate **80** may have therein a power line, in the form of power lines VPR**1** and VPR**2**, which is provided to supply a power to the semiconductor memory device **1000**. Therefore, a power line may be omitted in the first metal layer M**1**. The first metal layer M**1** may be electrically connected to the active contact AC and the upper wiring line W**1**.

[0043] A second metal layer M**2** may be provided in the third interlayer dielectric layer **130**. The second metal layer M**2** may include second wiring lines. The second wiring lines may be electrically connected to the first wiring lines of the first metal layer M**1**.

[0044] A third metal layer M**3** may be provided in the fourth interlayer dielectric layer **140**. The third metal layer M**3** may include third wiring lines. The third wiring lines may be electrically connected to the second wiring lines of the second metal layer M**2**.

[0045] A first support substrate **300** may be formed on the fourth interlayer dielectric layer **140**. The first support substrate **300** may be a silicon substrate, but embodiments of the present inventive concepts are not limited thereto.

[0046] A first lower dielectric layer **90** may be provided beneath the first substrate **100**. A lower substrate **80** may be provided beneath the first lower dielectric layer **90**. In other embodiments, the first lower dielectric layer **90** may be omitted. Dissimilarly, at least one lower interlayer dielectric layer may be formed between the first lower dielectric layer **90** and the lower substrate **80**.
[0047] The lower substrate **80** may be provided therein with first and second lower power lines VPR**1** and VPR**2**. The first and second lower power lines VPR**1** and VPR**2** may extend in parallel in a second direction D**2** (see FIG. **3**B). The peripheral region R may be positioned between the first lower power line VPR**1** and the second lower power line VPR**2**. The first and second lower power lines VPR**1** and VPR**2** may include at least one material selected from copper, molybdenum, tungsten, and ruthenium.
[0048] A power delivery network layer PDL may be provided in the lower substrate **80**. The power delivery network layer PDL may be provided beneath (in the vertical direction D**3**) the first and second lower power lines VPR**1** and VPR**2**. The power delivery network layer PDL may include a plurality of lower wiring lines electrically connected to the first and second lower power lines VPR**1** and VPR**2**.
[0049] A first lower power via PVI**1** may be provided between the power delivery network layer PDL and the first and second lower power lines VPR**1** and VPR**2**. The first lower power via PVI**1** may electrically connect the power delivery network layer PDL to the first and second lower power lines VPR**1** and VPR**2**.
[0050] The power delivery network layer PDL may apply a voltage to the first and second lower power lines VPR**1** and VPR**2**. For example, the power delivery network layer PDL may provide the first and second lower power lines VPR**1** and VPR**2** with at least one voltage selected from VSS, VDD, VBB, VBL, VEQ, VREFA, VINTY, VREFBB, VPP_DIV, VBB1_DIV, VTG_MON1, VREFY, VLANG, VINTA, VREF, VREFPP, VREFLANG, VREFLP, and VPP.
[0051] For example, the power delivery network layer PDL may include a wiring network for applying a source voltage VSS to the first lower power lines VPR**1**. The power delivery network layer PDL may include a wiring network for applying a drain voltage VDD to the second lower power line VPR**2**.
[0052] The upper structure US may include a backside through via BVI that vertically (i.e., in the D**3** direction) penetrates or extends through the deep device isolation pattern DTI. The backside through via BVI may vertically (i.e., in the D**3** direction) and electrically connect the upper wiring layer **111** on the first substrate **100** to the first and second lower power lines VPR**1** and VPR**2**. For example, a first backside through via BVI**1** may vertically (i.e., in the D**3** direction) and electrically connect the upper wiring line W**1** of the upper wiring layer **111** to the first lower power line VPR**1**. A second backside through via BVI**2** may vertically (i.e., in the D**3** direction) and electrically connect the upper wiring line W**1** of the upper wiring layer **111** to the second lower power line VPR**2**. The first and second backside through via BVI**1** and BVI**2** may each be electrically connected through the upper wiring line W**1** to the active contact AC.
[0053] The first and second backside through vias BVI**1** and BVI**2** may each have a first width WD**1** in the first direction D**1**. The first width WD**1** may decrease with increasing distance in the vertical direction D**3** from top surfaces of the first and second lower power lines VPR**1** and VPR**2**. The first and second backside through vias BVI**1** and BVI**2** may include a metallic material.
[0054] The lower structure LS will be further discussed in detail with reference to FIGS. **3**A and **4**.
[0055] The lower structure LS may include a second substrate **200**. The second substrate **200** may be a silicon substrate, but the present inventive concepts are not limited thereto.
[0056] A first lower interlayer dielectric layer **50** may be provided on the second substrate **200**. The first lower interlayer dielectric layer **50** may include a dielectric material. For example, the first lower interlayer dielectric layer **50** may include a plurality of dielectric layers.
[0057] A memory cell region **51** may be provided in the first lower interlayer dielectric layer **50**. For example, a vertical channel transistor DRAM may be provided on the second substrate **200**, but

the present inventive concepts are not limited thereto. The memory cell region **51** of FIG. **4** may correspond to a cross-sectional view taken along line A-A′ of FIG. **3**A.

[0058] The second substrate **200** may be provided thereon with a peripheral circuit structure PS and a cell array structure CS on the peripheral circuit structure PS. The peripheral circuit structure PS may include core circuits, a lower dielectric layer that covers the core circuits, lower contact plugs, and circuit lines.

[0059] The core circuits may include NMOS and PMOS transistors. The core circuits may be electrically connected through the circuit lines to bit lines BL and word lines WL, which will be discussed below.

[0060] The cell array structure CS may include bit lines BL, channel patterns CP, word lines WL, mold dielectric patterns IL, landing pads LP, and data storage patterns DSP.

[0061] On the second substrate **200**, the bit lines BL may extend in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. For example, the bit lines BL may include doped polysilicon, metal, conductive metal nitride, conductive metal silicide, conductive metal oxide, or any combination thereof. The bit lines BL may each be formed of a single or multiple layer. The bit lines BL may include a carbon-based two-dimensional material such as graphene, a carbon-based three-dimensional material such as carbon nano-tube, or any combination thereof.

[0062] The mold dielectric patterns IL may be provided on the bit lines BL. Each of the mold dielectric patterns IL may extend in the second direction D**2**, while running across the bit lines BL. The mold dielectric patterns IL may include, for example, one or more of silicon oxide, silicon nitride, silicon oxynitride, and low-k dielectrics.

[0063] The channel patterns CP may be positioned on the bit lines BL. The channel patterns CP may be spaced apart from each other in the second direction D**2** between the mold dielectric patterns IL. Each of the channel patterns CP may have a U-shaped cross-section. A width in the second direction D**2** of each of the channel patterns CP may be greater than a width in the second direction D**2** of each of the bit lines BL.

[0064] The channel patterns CP may include, for example, an oxide semiconductor. The oxide semiconductor may include In.sub.xGa.sub.yZn.sub.zO, In.sub.xGa.sub.ySi.sub.zO, In.sub.xSn.sub.yZn.sub.zO, In.sub.xZn.sub.yO, Zn.sub.xO, Zn.sub.xSn.sub.yO, Zn.sub.xO.sub.yN, Zr.sub.xZn.sub.ySn.sub.zO, Sn.sub.xO, Hf.sub.xIn.sub.yZn.sub.zO, Ga.sub.xZn.sub.ySn.sub.zO, Al.sub.xZn.sub.ySn.sub.zO, Yb.sub.xGa.sub.yZn.sub.zO, In.sub.xGa.sub.yO, or any combination thereof. In other embodiments, the channel patterns CP may include indium-gallium-zinc oxide (IGZO). The channel patterns CP may each be formed of a single layer including one oxide semiconductor or a multiple layer including different oxide semiconductors. The channel patterns CP may include an amorphous, crystalline, or polycrystalline oxide semiconductor. The channel patterns CP may each have bandgap energy greater than that of silicon. For example, the channel patterns CP may each have bandgap energy of about 1.5 eV to about 5.6 eV, and may exhibit desired channel performance when the bandgap energy is in a range of about 2.0 eV to about 4.0 eV.

[0065] The word lines WL may be positioned on the channel patterns CP. The word lines WL may extend in the second direction D**2**, while running across the bit lines BL and the channel patterns CP.

[0066] For example, the word lines WL may include doped polysilicon, metal, conductive metal nitride, conductive metal silicide, conductive metal oxide, or any combination thereof. The word lines WL may each be formed of a single layer or a multiple layer including different materials. The word lines WL may include a carbon-based two-dimensional material, such as graphene, a carbon-based three-dimensional material such as carbon nano-tube, or any combination thereof.

[0067] A first dielectric pattern **115** may be provided between a pair of word lines WL. Each of the first dielectric patterns **115** may be positioned between a corresponding pair of word lines WL.

Each of the first dielectric patterns **115** may extend along the second direction D**2**. The first dielectric patterns **115** may be spaced apart from each other in the first direction D**1**. The first dielectric patterns **115** and the mold dielectric patterns IL may be alternately arranged along the first direction D**1**.

[0068] Landing pads LP may be positioned on corresponding word lines WL. The landing pads LP may be in contact with the channel patterns CP and the mold dielectric patterns IL. Each of the landing pads LP may have a circular shape when viewed in plan, but the present inventive concepts are not limited thereto. For example, each of the landing pads LP may have an oval shape, a rectangular shape, a square shape, a rhombic shape, a hexagonal shape, or any other suitable shapes when viewed in plan. The landing pads LP may be spaced apart from each other in the first direction D**1** and the second direction D**2**. The landing pads LP may include, for example, doped polysilicon, metal, conductive metal nitride, conductive metal silicide, conductive metal oxide, or any combination thereof.

[0069] The data storage patterns DSP may be positioned on the landing pads LP. The data storage patterns DSP may be electrically connected through the landing pads LP to the channel patterns CP. When viewed in plan, each of the data storage patterns DSP may overlap a corresponding one of the landing pads LP. For example, the data storage patterns DSP may be spaced apart from each other in the first direction D**1** and the second direction D**2**.

[0070] For example, the data storage patterns DSP may each be a capacitor. In this case, the data storage patterns DSP may include a bottom electrode, a top electrode, and a dielectric layer between the bottom electrode and the top electrode. The bottom electrode may be in contact with the landing pad LP. In other embodiments, the data storage patterns DSP may each be a variable resistance pattern whose two resistance states can be switched due to an electrical pulse applied to a memory element. In this case, the data storage patterns DSP may include a phase-change material whose crystalline state is changed based on an amount of current, perovskite compounds, transition metal oxide, magnetic materials, ferromagnetic materials, or antiferromagnetic materials.

[0071] A plurality of wiring lines may be provided in the first lower interlayer dielectric layer **50**. For example, a first connection line **52** and a second connection line **53** may be provided in the first lower interlayer dielectric layer **50**. The first connection line **52** may be provided beneath (i.e., in the D**3** direction) and electrically connected to the data storage pattern DSP. The second connection line **53** may be provided on the first connection line **52** and may be electrically connected to the first connection line **52** through a first through via TVI**1**. The second connection line **53** may be electrically connected through a second through via TVI**2** to a lower wiring layer **61** which will be discussed below. In other embodiments, the connection lines **52** and **53** and the through vias TVI**1** and TVI**2** may be variously disposed. Dissimilarly, at least one wiring line may be additionally provided in the first lower interlayer dielectric layer **50**.

[0072] A second lower interlayer dielectric layer **60** may be formed on the memory cell region **51**. A lower wiring layer **61** may be formed in the second lower interlayer dielectric layer **60**. The lower wiring layer **61** may include a plurality of wiring lines. The data storage patterns DSP may be electrically connected to the lower wiring layer **61** through wiring lines in the first lower interlayer dielectric layer **50**. For example, the data storage patterns DSP may be electrically connected to the lower wiring layer **61** through the first connection line **52**, the first through via TVI**1**, the second connection line **53**, and the second through via TVI**2**.

[0073] A third lower interlayer dielectric layer **70** may be provided on the second lower interlayer dielectric layer **60**. The third lower interlayer dielectric layer **70** may include a dielectric material.

[0074] A bonding structure BD may be provided between the lower structure LS and the upper structure US. The bonding structure BD may be provided on a boundary between the lower structure LS and the upper structure US. A top surface of the bonding structure BD may be located at a higher level (i.e., in the D**3** direction) than that of the rear surface USa of the upper structure US. A bottom surface of the bonding structure BD may be located at a higher level (i.e., in the D**3**

direction) than that of the front surface LSa of the lower structure LS.
[0075] The bonding structure BD may include a first bonding pad **85** and a second bonding pad **75**. The first bonding pad **85** may be buried in a lower portion of the upper structure US. For example, the first bonding pad **85** may be buried in the lower substrate **80**. The second bonding pad **75** may be buried in an upper portion of the lower structure LS. For example, the second bonding pad **75** may be buried in the third lower interlayer dielectric layer **70** of the lower structure LS. The first bonding pad **85** may be exposed through the rear surface USa of the upper structure US. The second bonding pad **75** may be exposed through the front surface LSa of the lower structure LS.
[0076] The first bonding pad **85** and the second bonding pad **75** may include the same material. The first bonding pad **85** and the second bonding pad **75** may include a metallic material. For example, the first bonding pad **85** and the second bonding pad **75** may include copper (Cu). The first bonding pad **85** and the second bonding pad **75** may directly contact each other. The first bonding pad **85** and the second bonding pad **75** may be integrally bonded into a single unitary or monolithic piece. No interface may be present between the first bonding pad **85** and the second bonding pad **75**.
[0077] The first bonding pad **85** may electrically connect the power delivery network layer PDL to the lower power lines VPR**1** and VPR**2**. For example, the first bonding pad **85** may be vertically (i.e., D**3** direction) and electrically connected to the power delivery network layer PDL through a second lower power via PVI**2**. In this configuration, the first bonding pad **85** may be electrically connected to the lower power lines VPR**1** and VPR**2** through a first lower power via PVI**1**, the power delivery network layer PDL, and the second lower power via PVI**2**.
[0078] The second bonding pad **75** may be electrically connected to the lower wiring layer **61**. For example, the second bonding pad **75** may be electrically connected through a lower via **73** to wiring lines of the lower wiring layer **61**. In this configuration, the second bonding pad **75** may be electrically connected to the data storage patterns DSP through the lower wiring layer **61** and the connection lines **52** and **53** in the first lower interlayer dielectric layer **50**.
[0079] The bonding structure BD may electrically connect the upper structure US and the lower structure LS to each other. For example, the bonding structure BD may electrically connect the lower power lines VPR**1** and VPR**2** to the lower wiring layer **61**. The upper structure US may be electrically connected to the lower structure LS through the first and second bonding pads **85** and **75**.
[0080] Nitride patterns **71** and **81** may be provided on the boundary between the upper structure US and the lower structure LS. A first nitride pattern **81** may be provided on the rear surface USa of the upper structure US. For example, the first nitride pattern **81** may be provided on a bottom surface of the lower substrate **80**. A second nitride pattern **71** may be provided on the front surface LSa of the lower structure LS. For example, the second nitride pattern **71** may be provided on a top surface of the third lower interlayer dielectric layer **70**.
[0081] The first nitride pattern **81** may extend in the first direction D**1** along the bottom surface of the lower substrate **80**. The second nitride pattern **71** may extend in the first direction D**1** along the top surface of the third lower interlayer dielectric layer **70**. The first and second nitride patterns **81** and **71** may vertically (i.e., D**3** direction) overlap each other. The first and second nitride patterns **81** and **71** may bond the upper structure US and the lower structure LS to each other.
[0082] The first and second nitride patterns **81** and **71** may include nitride. For example, the first and second nitride patterns **81** and **71** may include silicon nitride (SiN). The first and second nitride patterns **81** and **71** may be disposed on one side of the bonding structure BD. A thickness of each of the first and second nitride patterns **81** and **71** may be less than that of the bonding structure BD.
[0083] According to some embodiments of the present inventive concepts, the upper structure US and the lower structure LS may be bonded to each other. A hybrid bonding manner may be employed to bond the upper structure US and the lower structure LS to each other. The term “hybrid bonding” may denote that two components including the same material are merged at an

interface therebetween. As the first bonding pad **85** and the second bonding pad **75** are bonded into a single unitary piece, the upper structure US and the lower structure LS may be stably bonded to each other.

[0084] In addition, as power lines of the upper structure US are buried in a lower portion of the first substrate **100**, there may be a reduction in resistance occurring between wiring lines of the upper structure US. The lower power lines VPR**1** and VPR**2** of the upper structure US may be electrically connected to the bonding structure BD. Thus, the upper structure US and the lower structure LS are stably combined through the bonding structure BD, and simultaneously, the bonding structure BD and power lines may be electrically connected to reduce resistance and to increase the degree of wiring freedom. In conclusion, the semiconductor memory device may increase in electrical properties.

[0085] FIGS. **5**A to **5**E and **6**A to **6**D are diagrams illustrating a method of fabricating a semiconductor memory device according to some embodiments of the present inventive concepts. FIGS. **5**A to **5**E are cross-sectional diagrams illustrating a method of manufacturing an upper structure US according to some embodiments of the present inventive concepts. FIGS. **6**A to **6**D are cross-sectional diagrams illustrating a method of manufacturing a lower structure LS according to some embodiments of the present inventive concepts.

[0086] Referring to FIG. **5**A, a peripheral region R may be formed on a first substrate **100**. In other embodiments, a cell region may be formed on the first substrate **100**. A deep device isolation pattern DTI may be formed on the first substrate **100**. The formation of the deep device isolation pattern DTI may include forming a hardmask pattern on the first substrate **100**, using the hardmask pattern as a mask to etch the first substrate **100** to form a deep device trench DTR, and at least partially filling the deep device trench DTR with a dielectric material.

[0087] An isolation pattern STI may be formed on the first substrate **100**. A maximum vertical (i.e., D**3** direction) depth of the isolation pattern STI may be less than that of the deep device isolation pattern DTI. The formation of the isolation pattern STI may include forming a hardmask pattern on the first substrate **100**, using the hardmask pattern as a mask to an upper portion of the first substrate **100** to form a trench, and at least partially filling the trench with a dielectric material.

[0088] The isolation pattern STI may define an active pattern AP. A transistor TR and source/drain patterns SD may be formed on the active pattern AP. The transistor TR may correspond to the selection element TR of FIG. **1**. The source/drain patterns SD may be formed on opposite sides of the transistor TR. The formation of the source/drain pattern SD may include implanting impurities into an upper portion of the active pattern AP.

[0089] A first interlayer dielectric layer **110** may be formed on the first substrate **100**. An upper wiring layer **111** may be formed in the first interlayer dielectric layer **110**. The upper wiring layer **111** may include active contacts AC, upper wiring lines W**1**, and upper vias VI. Each of the active contacts AC may be vertically (i.e., D**3** direction) and electrically connected to a corresponding source/drain pattern SD. The upper vias VI may be provided beneath the upper wiring lines W**1**. The upper wiring lines W**1** may be connected through the upper vias VI to the active contacts AC.

[0090] Referring to FIG. **5**B, a second interlayer dielectric layer **120** may be formed on the first interlayer dielectric layer **110**. The second interlayer dielectric layer **120** may include a silicon oxide layer. A first metal layer M**1** may be formed in the second interlayer dielectric layer **120**. The first metal layer M**1** may be electrically connected to at least one of the active contacts AC and the upper wiring lines W**1**.

[0091] A third interlayer dielectric layer **130** may be formed on the second interlayer dielectric layer **120**. A second metal layer M**2** may be formed in the third interlayer dielectric layer **130**. A fourth interlayer dielectric layer **140** may be formed on the third interlayer dielectric layer **130**. A third metal layer M**3** may be formed in the fourth interlayer dielectric layer **140**. A first support substrate **300** may be formed on the fourth interlayer dielectric layer **140**. The first support substrate **300** may be a silicon substrate, but the present inventive concepts are not limited thereto.

[0092] Referring to FIG. **5**C, after a back-end-of-line (BEOL) process is completed, the first substrate **100** may be overturned to expose a bottom surface of the first substrate **100**. A silicon grinding process and/or a chemical mechanical polishing (CMP) process may be employed to reduce a thickness of the first substrate **100**. For example, a thickness of the first substrate **100** depicted in FIG. **5**B may be less than that of the first substrate **100** depicted in FIG. **5**C. The reduction in thickness of the first substrate **100** may at least partially expose a bottom surface of the deep device isolation pattern DTI. The reduction in thickness of the first substrate **100** may remove a portion of the deep device isolation pattern DTI.

[0093] Referring to FIG. **5**D, a first lower dielectric layer **90** may be formed on the bottom surface of the first substrate **100**. A backside contact hole BCH may be formed in the first lower dielectric layer **90**, the first substrate **100**, and the first interlayer dielectric layer **110**. The backside contact hole BCH may penetrate or extend through the first lower dielectric layer **90** and the deep device isolation pattern DTI. The backside contact hole BCH may at least partially expose the upper wiring line W**1** in the first interlayer dielectric layer **110**. A width in a first direction D**1** of the backside contact hole BCH may increase with increasing distance in a vertical (i.e., D**3** direction) direction D**3** from an exposed top surface of the upper wiring line W**1**.

[0094] The formation of the backside contact hole BCH may include, for example, forming a mask pattern on the first lower dielectric layer **90**, using the mask pattern as a mask to etch the first lower dielectric layer **90**, the deep device isolation pattern DTI, and the first interlayer dielectric layer **110**.

[0095] Referring to FIG. **5**E, the backside contact hole BCH may be at least partially filled with a metallic material to form a backside through via BVI. For example, the backside through via BVI may include at least one metallic material selected from aluminum, copper, tungsten, molybdenum, ruthenium, and cobalt.

[0096] Referring to FIG. **4**, a lower substrate **80** may be formed on the first lower dielectric layer **90**. The lower substrate **80** may include a dielectric material. First and second lower power lines VPR**1** and VPR**2** may be formed in the lower substrate **80**. The first and second lower power lines VPR**1** and VPR**2** may be connected to the backside through via BVI.

[0097] A first lower power via PVI**1** and a power delivery network layer PDL may be formed on the first and second lower power lines VPR**1** and VPR**2**. The power delivery network layer PDL may be configured to apply a voltage to the first and second lower power lines VPR**1** and VPR**2**. For example, the power delivery network layer PDL may apply a source voltage VSS or a drain voltage VDD to the first and second lower power lines VPR**1** and VPR**2**.

[0098] A first nitride pattern **81**, a second lower power via PVI**2**, and a first bonding pad **85** may be sequentially formed in the lower substrate **80**. The formation of the first nitride pattern **81** may include conformally forming a nitride layer on a top surface of the lower substrate **80**, and performing a chemical mechanical polishing (CMP) process to reduce a thickness of the nitride layer.

[0099] The formation of the first bonding pad **85** may include forming a mask pattern on the top surface of the lower substrate **80**, using the mask pattern as a mask to etch an upper portion of the lower substrate **80**, and at least partially filling the etched upper portion of the lower substrate **80** with a metallic material (e.g., copper (Cu)).

[0100] Referring to FIG. **6**A, a peripheral circuit structure PS including core circuits may be formed on a first preliminary substrate **10**.

[0101] Bit lines BL may be formed on the peripheral circuit structure PS. Mold dielectric patterns IL may be formed on the bit lines BL. The mold dielectric patterns IL may extend in a second direction D**2** and may be spaced apart from each other in the first direction D**1**. The mold dielectric patterns IL may at least partially expose portions of the bit lines BL. The mold dielectric patterns IL may include, for example, one or more of silicon oxide, silicon nitride, silicon oxynitride, and low-k dielectrics.

[0102] A channel layer may be formed to at least partially cover the mold dielectric patterns IL. The channel layer may be formed to have a uniform thickness. The channel layers may be in contact with portions of the bit lines BL. The channel layer may be formed by using at least one process selected from physical vapor deposition (PVD), thermal chemical deposition process (thermal CVD), low pressure chemical vapor deposition (LPCVD), plasma enhanced chemical vapor deposition (PECVD), and atomic layer deposition (ALD). For example, the channel layer may include a semiconductor material, an oxide semiconductor material, or a two-dimensional semiconductor material, or may include silicon, germanium, silicon-germanium, or indium gallium zinc oxide (IGZO).
[0103] An etching process may be performed on the channel layer. A portion of the channel layer may be removed such that the channel layer may be formed into channel patterns CP. After the formation of the channel patterns CP, word lines WL may be formed on the channel patterns CP. The word lines WL may have their top surfaces lower (i.e., lower in the D**3** direction) than those of the channel patterns CP.
[0104] Afterwards, first dielectric patterns **115** may be formed between a pair of word lines WL positioned on one channel pattern CP. Each of the first dielectric patterns **115** may at least partially fill between a pair of word lines WL.
[0105] Landing pads LP may be formed on the channel pattern CP and the word line WL. The landing pads LP may be spaced apart from each other in the first direction D**1** and the second direction D**2**. Data storage patterns DSP may be correspondingly formed on the landing pads LP. When the data storage patterns DSP include capacitors, bottom electrodes, a capacitor dielectric layer, and a top electrode may be sequentially formed. The bottom electrodes may be electrically connected to corresponding landing pads LP.
[0106] A first lower interlayer dielectric layer **50** may be formed on the first preliminary substrate **10**. A plurality of wiring lines may be formed in the first lower interlayer dielectric layer **50**. For example, there may be formed wiring lines electrically connected to the data storage pattern DSP. For example, a first connection line **52**, a second connection line **53**, and a first through via TVI**1** may be formed on the first preliminary substrate **10**. A photoresist process may be employed to form the first connection line **52**, the second connection line **53**, and the first through via TVI**1**. For another example, a plurality of wiring lines may be variously formed in the first lower interlayer dielectric layer **50**.
[0107] Referring to FIG. **6**B, a second substrate **200** may be formed on the first lower interlayer dielectric layer **50**. The second substrate **200** may be a silicon substrate, but the present inventive concepts are not limited thereto.
[0108] Referring to FIG. **6**C, the first preliminary substrate **10** may be overturned to at least partially expose a bottom surface of the first preliminary substrate **10**. A silicon grinding process and/or a chemical mechanical polishing (CMP) process may be employed to remove the first preliminary substrate **10**. The first preliminary substrate **10** may be removed to expose a top surface of the first lower interlayer dielectric layer **50**.
[0109] Referring to FIG. **6**D, a second lower interlayer dielectric layer **60** may be formed on the first lower interlayer dielectric layer **50**. A second through via TVI**2** may be formed in the first lower interlayer dielectric layer **50**. A lower wiring layer **61** may be formed in the second lower interlayer dielectric layer **60**. The second through via TVI**2** may electrically connect the lower wiring layer **61** to the second connection line **53**.
[0110] A third lower interlayer dielectric layer **70** may be formed on the second lower interlayer dielectric layer **60**. A second nitride pattern **71**, a lower via **73**, and a second bonding pad **75** may be formed in the third lower interlayer dielectric layer **70**. The formation of the second nitride pattern **71** may include forming a nitride layer on a top surface of the third lower interlayer dielectric layer **70** and performing a chemical mechanical polishing (CMP) process to reduce a thickness of the nitride layer.

[0111] The formation of the second bonding pad **75** may include forming a mask pattern on the third lower interlayer dielectric layer **70**, using the mask pattern as a mask to etch an upper portion of the third lower interlayer dielectric layer **70**, and at least partially filling the etched upper portion of the third lower interlayer dielectric layer **70** with a metallic material. The second bonding pad **75** may be electrically connected through the lower via **73** to the lower wiring layer **61**.
[0112] Referring again to FIG. **4**, the upper structure US and the lower structure LS may vertically (i.e., D**3** direction) overlap to bond to each other. A rear surface USa of the upper structure US may be in contact with a front surface LSa of the lower structure LS. The first nitride pattern **81** of the upper structure US may be in contact with the second nitride pattern **71** of the lower structure LS. The first and second nitride patterns **81** and **71** may bond the upper structure US and the lower structure LS to each other. The first bonding pad **85** and the second bonding pad **75** may vertically (i.e., D**3** direction) overlap to contact each other.
[0113] An annealing process may be performed on the upper structure US and the lower structure LS. The annealing process may direct bond the first bonding pad **85** and the second bonding pad **75** to each other. For example, the first bonding pad **85** and the second bonding pad **75** may form a single unitary or monolithic piece.
[0114] The first bonding pad **85** and the second bonding pad **75** may be formed of the same material (e.g., copper (Cu)). The first bonding pad **85** and the second bonding pad **75** may be combined due to an intermetallic hybrid bonding process caused by surface activation at an interface between the first bonding pad **85** and the second bonding pad **75** that are in contact with each other. When the first bonding pad **85** and the second bonding pad **75** are bonded to each other, no perfect incorporation may be achieved.
[0115] FIG. **7** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts. In the embodiment that follows, a detailed description of technical features repetitive to those discussed above with reference to FIG. **4** will be omitted, and a difference thereof will be discussed in detail.
[0116] Referring to FIG. **7**, the upper structure US may be provided on the lower structure LS. The lower structure LS may include a power wiring layer PDN. The lower wiring layer PDN may be provided on the lower wiring layer **61**. The power wiring layer PDN may include a first power wiring dielectric layer PL**1** and a second power wiring dielectric layer PL**2**. The second power wiring dielectric layer PL**2** may be provided on the first power wiring dielectric layer PL**1**. A first bonding pattern PBa may be provided between the first power wiring dielectric layer PL**1** and the second power wiring dielectric layer PL**2**. The first bonding pattern PBa may include the same material as that of the first and second nitride patterns **71** and **81** discussed above. The first bonding pattern PBa may bond the first power wiring dielectric layer PL**1** and the second power wiring dielectric layer PL**2** to each other.
[0117] The power delivery network layer PDL may be provided in the first power wiring dielectric layer PL**1**. The first lower power via PVI**1** may be provided beneath the power delivery network layer PDL. The power delivery network layer PDL may be electrically connected through the first lower power via PVI**1** to the lower wiring layer **61**.
[0118] The first and second lower power lines VPR**1** and VPR**2** may be provided in the second power wiring dielectric layer PL**2**. The second lower power via PVI**2** may be provided beneath (i.e., D**3** direction) the first and second lower power lines VPR**1** and VPR**2**. The second lower power via PVI**2** may connect the first lower power line VPR**1** to the power delivery network layer PDL. The second lower power via PVI**2** may connect the second lower power line VPR**2** to the power delivery network layer PDL.
[0119] The lower substrate **80** may be omitted in the upper structure US. The first lower dielectric layer **90** of the upper structure US may be provided on the second power wiring dielectric layer PL**2** of the upper structure US.
[0120] A second bonding pattern PBb and a third bonding pattern PBc may be interposed between

the second lower wiring dielectric layer PL**2** and the first lower dielectric layer **90**. The second bonding pattern PBb may be provided on the front surface LSa of the lower structure LS. The third bonding pattern PBc may be provided on the rear surface USa of the upper structure US. The second and third bonding patterns PBb and PBc may be the same as the first and second nitride patterns **81** and **71** discussed above.

[0121] The first and second backside through vias BVI**1** and BVI**2** may be provided on the first substrate **100** of the upper structure US. The first and second backside through vias BVI**1** and BVI**2** may each have a second width WD**2** in the first direction D**1**. The second width WD**2** may increase with increasing distance in the vertical direction D**3** from the rear surface USa of the upper structure US.

[0122] According to the present inventive concepts, the power wiring layer PDN may be disposed in the lower structure LS. Thus, there may be a reduction in resistance between wiring lines in the lower structure LS. In other embodiments, the bonding structure BD of FIG. **4** may be disposed on the boundary between the upper structure US and the lower structure LS.

[0123] FIGS. **8**A, **8**B, **9**A, **9**B, **10**, **11**, and **12** are diagrams illustrating a method of fabricating a semiconductor memory device according to some embodiments of the present inventive concepts. In the embodiment that follows, a detailed description of technical features repetitive to those discussed with reference to FIGS. **5**A to **6**D will be omitted, and a difference thereof will be discussed in detail.

[0124] Referring to FIG. **8**A, a peripheral region R may be formed on a first substrate **100**. In other embodiments, a cell region may be formed on the first substrate **100**. A deep device isolation pattern DTI, an isolation pattern STI, a transistor TR, and source/drain patterns SD may be formed on the first substrate **100**. A first interlayer dielectric layer **110** may be formed on the first substrate **100**. Active contacts AC may be formed in the first interlayer dielectric layer **110**.

[0125] Referring to FIG. **8**B, the first substrate **100** may be overturned to at least partially expose a bottom surface of the first substrate **100**. A silicon grinding process and/or a chemical mechanical polishing (CMP) process may be employed to reduce a thickness of the first substrate **100**. The reduction in thickness of the first substrate **100** may at least partially expose a bottom surface of the deep device isolation pattern DTI.

[0126] A first lower dielectric layer **90** may be formed on the bottom surface of the first substrate **100**. A first lower dielectric layer **90** may be formed on the first lower dielectric layer **90**. The formation of the third bonding pattern PBc may include conformally forming a nitride layer on the first lower dielectric layer **90** and performing a chemical mechanical polishing (CMP) process to reduce a thickness of the nitride layer. Through the processes mentioned above, an upper structure US may be formed.

[0127] Referring to FIG. **9**A, a first lower interlayer dielectric layer **50** may be formed on a second substrate **200**. A memory cell region **51**, connection lines **52** and **53**, and through vias TVI**1** and TVI**2** may be formed in the first lower interlayer dielectric layer **50**. A second lower interlayer dielectric layer **60** may be formed on the first lower interlayer dielectric layer **50**. A lower wiring layer **61** may be formed in the second lower interlayer dielectric layer **60**.

[0128] A first power wiring dielectric layer PL**1** may be formed on the second lower interlayer dielectric layer **60**. A second lower power via PVI**2** and a power delivery network layer PDL may be sequentially formed in the first power wiring dielectric layer PL**1**. A first bonding pattern PBa may be formed on the first power wiring dielectric layer PL**1**. The formation of the first bonding pattern PBa may include conformally forming a nitride layer on the first lower dielectric layer **90** and performing a chemical mechanical polishing (CMP) process to reduce a thickness of the nitride layer.

[0129] Referring to FIG. **9**B, a second power wiring dielectric layer PL**2** may be formed on the first power wiring dielectric layer PL**1**. The first power wiring dielectric layer PL**1** and the second power wiring dielectric layer PL**2** may be combined through the first bonding pattern PBa. A first lower power via PVI**1** and first and second lower power lines VPR**1** and VPR**2** may be sequentially formed in the second power wiring dielectric layer PL**2**. The first lower power via PVI**1** may penetrate the first bonding pattern PBa.

[0130] A second bonding pattern PBb may be formed on the second power wiring dielectric layer PL**2**. The formation of the second bonding pattern PBb may include conformally forming a nitride layer on the first lower dielectric layer **90** and performing a chemical mechanical polishing (CMP) process to reduce a thickness of the nitride layer. Through the processes mentioned above, a lower structure LS may be formed.

[0131] Referring to FIG. **10**, the upper structure US and the lower structure LS may be bonded to each other. A rear surface USa of the upper structure US may be in contact with a front surface LSa of the lower structure LS. The upper structure US and the lower structure LS may be combined through the second bonding pattern PBb and the third bonding pattern PBc.

[0132] A backside contact hole BCH may be formed in the upper structure US. The backside contact hole BCH may penetrate or extend through the first interlayer dielectric layer **110**, the first substrate **100**, and the first lower dielectric layer **90** to at least partially expose the first and second lower power lines VPR**1** and VPR**2** of the lower structure LS. The backside contact hole BCH may penetrate or extend through the second and third bonding patterns PBb and PBc. A width in the first direction D**1** of the backside contact hole BCH may decrease with increasing distance in the vertical direction D**3** from a top surface of the first interlayer dielectric layer **110**.

[0133] The formation of the backside contact hole BCH may include, for example, forming a mask pattern on the first interlayer dielectric layer **110** and using the mask pattern as a mask to etch the first interlayer dielectric layer **110**, the first substrate **100**, and the first lower dielectric layer **90**.

[0134] Referring to FIG. **11**, the backside contact hole BCH may be at least partially filled with a metallic material to form a backside through via BVI. For example, the backside through via BVI may include at least one metallic material selected from aluminum, copper, tungsten, molybdenum, ruthenium, and cobalt.

[0135] Referring to FIG. **12**, an upper wiring line W**1** and upper vias VI may be formed in the first interlayer dielectric layer **110**. The transistor TR and the source/drain patterns SD may be electrically connected through the active contact AC to the wiring line W**1**. The backside through via BVI may vertically and electrically connect the upper wiring line W**1** to the first and second lower power lines VPR**1** and VPR**2**. The active contact AC may be electrically connected through the upper wiring line W**1** to the backside through via BVI.

[0136] Referring to FIG. **7**, second, third, and fourth interlayer dielectric layers **120**, **130**, and **140** may be formed on the first interlayer dielectric layer **110**. A first metal layer M**1** may be formed in the second interlayer dielectric layer **120**. A second metal layer M**2** may be formed in the third interlayer dielectric layer **130**. A third metal layer M**3** may be formed in the fourth interlayer dielectric layer **140**. A first support substrate **300** may be formed on the fourth interlayer dielectric layer **140**. In other embodiments, the first support substrate **300** may be omitted.

[0137] FIG. **13** is a cross-sectional view illustrating a semiconductor memory device according to some embodiments of the present inventive concepts. In the embodiment that follows, a detailed description of technical features repetitive to those discussed above with reference to FIG. **4** will be omitted, and a difference thereof will be discussed in detail.

[0138] Referring to FIG. **13**, the channel pattern CP may be provided on the bit line BL. The channel pattern CP may extend in the vertical direction D**3** from a top surface of the bit line BL. The word line WL may be provided between a pair of channel patterns CP that are adjacent to each other. The word lines WL may be spaced apart from each other in the second direction D**2**. The mold dielectric patterns IL may be provided between a pair of channel patterns CP that are adjacent to each other. The mold dielectric patterns IL may be spaced apart from each other in the second direction D**2**. The word lines WL and the mold dielectric patterns IL may be alternately provided between the channel patterns CP. The bit line BL may extend in the first direction D**1**, and the word

line WL may extend in the second direction D**2**.
[0139] In a semiconductor memory device according to some embodiments of the present inventive concepts, a lower structure and an upper structure may be combined through a bonding structure. The upper structure may include a lower power line buried in a lower portion of a substrate, and the lower power line may be electrically connected to the bonding structure. The upper structure may be electrically connected through the bonding structure to the lower structure.
[0140] The bonding structure may include a first bonding pad buried in a lower portion of the upper structure and a second bonding pad buried in an upper portion of the lower structure. The first bonding pad and the second bonding pad may include the same material and may constitute a single unitary or monolithic piece. Thus, the lower structure and the upper structure may be connected to each other, and at the same time, the lower power line of the upper structure and the bonding structure may be electrically connected, which may result in an increase in the degree of wiring freedom. In conclusion, the semiconductor memory device may increase in desired electrical properties.
[0141] Although the present invention has been described in connection with the embodiments of the present inventive concepts illustrated in the accompanying drawings, it will be understood to those skilled in the art that various changes and modifications may be made without departing from the technical spirit and essential feature of the present inventive concepts. It therefore will be understood that the embodiments described above are just illustrative but not limitative in all aspects.

Claims

1. A semiconductor memory device, comprising: a lower structure; and an upper structure on the lower structure, wherein the upper structure includes: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a first bonding pad between the lower power line and the lower structure, wherein the lower power line and the first bonding pad are electrically connected to each other, and wherein the upper structure is electrically connected through the first bonding pad to the lower structure.
2. The device of claim 1, wherein the lower structure includes a second bonding pad, wherein the first bonding pad and the second bonding pad directly contact each other.
3. The device of claim 2, wherein the lower structure further includes: a second substrate; a data storage pattern on the second substrate; a landing pad on the data storage pattern; a channel pattern on the landing pad, wherein the landing pad electrically connects the channel pattern and the data storage pattern to each other; a word line adjacent to the channel pattern; and a bit line on the channel pattern, wherein the bit line extends in a first direction, and wherein the word line extends in a second direction that intersects the first direction.
4. The device of claim 3, further comprising a lower wiring line between the bit line and the second bonding pad, wherein the lower wiring line is electrically connected to the second bonding pad.
5. The device of claim 1, further comprising a deep device isolation pattern that extends through the first substrate.
6. The device of claim 5, further comprising a backside through via that extends through the deep device isolation pattern, wherein the backside through via electrically connects the upper wiring line and the lower power line to each other.
7. The device of claim 6, wherein a width of the backside through via decreases with distance in a direction perpendicular to a top surface of the lower power line extending away from the lower structure.
8. The device of claim 6, further comprising: a transistor on the first substrate; a plurality of source/drain patterns on opposite sides of the transistor; and an active contact electrically connected to each of the source/drain patterns, wherein the active contact is electrically connected

to the backside through via.

9. The device of claim 1, further comprising a power delivery network layer between the lower power line and the lower structure, wherein the power delivery network layer is configured to apply a voltage to the lower power line.

10. The device of claim 1, further comprising a nitride pattern between the upper structure and the lower structure.

11. A semiconductor memory device, comprising: a lower structure; and an upper structure on the lower structure, wherein the upper structure includes: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a first bonding pad between the lower power line and the lower structure, wherein the lower structure includes: a second substrate that includes a cell region; and a second bonding pad on the cell region, wherein the first bonding pad is exposed through a rear surface of the upper structure, wherein the second bonding pad is exposed through a front surface of the lower structure, wherein the front surface of the lower structure is in contact with the rear surface of the upper structure, and wherein the first bonding pad and the second bonding pad directly contact each other.

12. The device of claim 11, further comprising a backside through via that extends through the first substrate and electrically connects the upper wiring line to the lower power line.

13. The device of claim 11, wherein the first bonding pad is electrically connected to the lower power line, and the second bonding pad is electrically connected to a lower wiring line on the cell region.

14. The device of claim 11, wherein the cell region of the lower structure includes: a data storage pattern on the second substrate; a landing pad on the data storage pattern; a channel pattern on the landing pad, wherein the landing pad electrically connects the channel pattern and the data storage pattern to each other; a word line adjacent to the channel pattern; and a bit line on the channel pattern, wherein the bit line extends in a first direction, and wherein the word line extends in a second direction that intersects the first direction.

15. The device of claim 11, further comprising a power delivery network layer between the lower power line and the lower structure, wherein the power delivery network layer is configured to apply a voltage to the lower power line.

16. A semiconductor memory device, comprising: a lower structure; an upper structure on the lower structure; and a bonding structure between the lower structure and the upper structure, wherein the upper structure includes: a first substrate; an upper wiring line on the first substrate; a lower power line in a lower portion of the first substrate; and a backside through via that extends through the first substrate and connects the upper wiring line to the lower power line, wherein the lower structure includes: a second substrate that includes a cell region; and a lower wiring layer on the cell region, wherein the bonding structure includes: a first part in a lower portion of the upper structure; and a second part in an upper portion of the lower structure, wherein the bonding structure electrically connects the lower power line and the lower wiring layer to each other.

17. The device of claim 16, further comprising: a transistor on the first substrate; a plurality of source/drain patterns on opposite sides of the transistor; and an active contact electrically connected to each of the source/drain patterns, wherein the active contact is electrically connected to the backside through via.

18. The device of claim 16, wherein at least one selected from the upper structure and the lower structure further includes: a data storage pattern on the second substrate; a landing pad on the data storage pattern; a channel pattern on the landing pad, wherein the landing pad electrically connects the channel pattern and the data storage pattern to each other; a word line adjacent to the channel pattern; and a bit line on the channel pattern, wherein the bit line extends in a first direction, and wherein the word line extends in a second direction that intersects the first direction.

19. The device of claim 16, further comprising a power delivery network layer between the lower power line and the lower structure, wherein the power delivery network layer is configured to apply a voltage to the lower power line.

20. The device of claim 16, further comprising a deep device isolation pattern that extends through the first substrate.